

# Condensation

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Video: <https://www.youtube.com/watch?v=odsHleu80x8>

## 1 Prerequisites

1. We use probability on finite sets, conditional probability, and elementary proofs.
2. All random variables in this sheet have finite ranges.
3. All logarithms have base two.
4. We use conditional entropy, conditional independence, and the distinction between an individual latent and a collection of latents.

### What you'll learn

#### Condensation

- Construct a latent model and distinguish LVM from reconstruction.
- Explain what the Markov condition excludes, using a concrete example.
- Prove correspondence between collections of latents, first exactly and then with explicit error terms.
- Distinguish the correspondence theorem from claims about learning, computationally finding a translation, or choosing observables.

#### A hierarchical clustering example

- Distinguish LVM, where the latents assigned to an observation determine it, from reconstruction, where an observation recovers those latents.
- Compare the information charged by different latent models for the same query.

## Condensation in context

- Relate condensation to sufficient statistics, common information, computational mechanics, and graphical models.
- Separate informational correspondence from finding and using a translation.
- Identify which additional claims would connect correspondence to AI safety.

## How to use this sheet

Read Sections 2.1 and 2.2 after the first lecture, then work on Exercises 2.1 to 2.3. After the second lecture, read Section 2.3 and complete Exercises 2.4 to 2.8. These eight exercises form the core route. Exercises 2.9 and 2.10 are optional extensions.

### 1.1 Information theory reference

For a finite-valued variable  $U$ ,

$$H(U) = - \sum_u p(u) \log p(u), \quad H(U \mid V) = \sum_v p(v) H(U \mid V = v),$$

with  $0 \log 0 = 0$ . Conditional entropy is nonnegative. It is zero precisely when  $U$  is a function of  $V$  almost surely: values on probability-zero events do not matter. The chain rule and conditioning inequality are

$$H(U, V \mid W) = H(U \mid W) + H(V \mid U, W), \quad H(U \mid V, W) \leq H(U \mid W).$$

Conditional mutual information is

$$I(U; V \mid W) = H(U \mid W) - H(U \mid V, W) \geq 0.$$

It is symmetric in  $U, V$  and is zero precisely when  $U, V$  are conditionally independent given  $W$ . Omitting  $W$  gives mutual information  $I(U; V)$ . Entropy of a tuple means joint entropy, not the sum of its marginal entropies. An empty tuple is constant and has zero entropy. For probability distributions  $p, q$  on a finite set,

$$D_{\text{KL}}(p \parallel q) = \sum_u p(u) \log \frac{p(u)}{q(u)} \geq 0.$$

This last quantity appears only in the optional score discussion.

## 2 Condensation

### 2.1 A family of questions

Fix jointly distributed observables  $X_1, \dots, X_n$ . We look for latent variables, each assigned to a set of observations, such that the latents assigned to an observation determine it. We will give conditions under which corresponding latent collections in two

such models determine one another, and bound the remaining conditional entropy when the conditions hold approximately. We take the joint distributions as given.

### Remark (Choosing the observables)

One can write  $X_i = f_i(S)$  for an underlying variable  $S$  representing sense data or a physical state. This introduces no restriction:  $S$  could simply be the tuple of all observations. An observable distinguishes outcomes by partitioning them into its possible answers; random-variable values label those parts. Information quantities do not depend on bijective relabeling of answers. The choice of which observables to include, and their joint law, remains part of the model.

**Example 2.1** (Overlapping observations). Let  $S, P, Q$  be independent fair bits and set

$$X_1 = (S, P), \quad X_2 = (S, Q), \quad X_3 = (P, Q).$$

Use a latent  $Y_{\{1,2\}} = S$  for observations 1 and 2, a latent  $Y_{\{1,3\}} = P$  for observations 1 and 3, and a latent  $Y_{\{2,3\}} = Q$  for observations 2 and 3. Each observation is determined by the two latents assigned to it. Each of these latents can also be recovered from either observation to which it is assigned.

#### 2.1.1 Addresses and collections

Let  $I = \{1, \dots, n\}$  and write  $\mathcal{P}^+(I)$  for its nonempty subsets. An *address*  $A \in \mathcal{P}^+(I)$  specifies which observations receive a latent  $Y_A$ . The collection of addresses is a partially ordered set under inclusion. We use one slot for each nonempty subset, setting unused slots to constants.

**Definition 2.2** (Latent model). A latent model of  $X = (X_i)_{i \in I}$  consists of finite-valued variables  $Y = (Y_A)_{A \in \mathcal{P}^+(I)}$  jointly distributed with  $X$ , such that

$$H(X_i \mid Y_{\{A \in \mathcal{P}^+(I) : i \in A\}}) = 0 \quad \text{for every } i \in I. \quad (1)$$

Here  $Y_{\mathcal{F}} = (Y_A)_{A \in \mathcal{F}}$  for a family  $\mathcal{F} \subseteq \mathcal{P}^+(I)$ .

We call (1) the *latent variable model condition*, or *LVM*. The contributing latents, taken together, determine the observation. This is not a claim that an agent knows their realized values. Some slots may represent residual noise. Also, an address need not be minimal: being assigned to an observation does not assert that a slot makes a nonredundant or causal contribution to it.

For nonempty  $A \subseteq I$  and  $B \subseteq I$ , define two different families:

$$\succeq A = \{C \in \mathcal{P}^+(I) : A \subseteq C\}, \quad \mathcal{J}_B = \{C \in \mathcal{P}^+(I) : C \cap B \neq \emptyset\}.$$

The tuple  $Y_{\succeq A}$  is the *tower above*  $A$ . The tuple  $Y_{\mathcal{J}_B}$  contains all latents contributing to at least one observation in  $B$ , and therefore determines  $X_B = (X_i)_{i \in B}$ . In particular

$\supseteq\{i\} = \mathcal{J}_{\{i\}}$  is the family of addresses contributing to  $X_i$ , so (1) reads  $H(X_i | Y_{\supseteq\{i\}}) = 0$ . For a larger  $A$ ,  $\supseteq A$  and  $\mathcal{J}_A$  generally differ.

**Exercise 2.1 (Overlapping bits; 15 minutes).** (★) Use Example 2.1, with all unmentioned slots constant.

- (a) Compute  $H(X_1)$ ,  $H(X_1, X_2, X_3)$ , and  $I(X_1; X_2)$ .
- (b) List  $\supseteq\{1\}$ ,  $\supseteq\{2\}$ , and  $\supseteq\{1, 2\}$ , including constant slots. Which tuple determines  $X_1$ ? Which of the three is  $\supseteq\{1\} \cap \supseteq\{2\}$ ?
- (c) Give explicit functions recovering  $X_1$  from its contributing latents and recovering  $Y_{\{1,2\}}$  from  $X_1$ .
- (d) Does the address structure have to be a tree? Identify two overlapping addresses in this example neither of which contains the other.

### Solution to Exercise 2.1

The independent fair bits are  $S, P, Q$ , and  $X_1 = (S, P)$ ,  $X_2 = (S, Q)$ ,  $X_3 = (P, Q)$ .

- (a)  $H(X_1) = 2$ . The tuple of all observations determines  $(S, P, Q)$  and is determined by it, so  $H(X_1, X_2, X_3) = 3$ . The pair  $(X_1, X_2)$  already determines all three bits, hence

$$I(X_1; X_2) = H(X_1) + H(X_2) - H(X_1, X_2) = 2 + 2 - 3 = 1.$$

Its shared bit is  $S$ .

- (b) The families, including constant slots, are

$$\begin{aligned}\supseteq\{1\} &= \{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}, \\ \supseteq\{2\} &= \{\{2\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}, \\ \supseteq\{1, 2\} &= \{\{1, 2\}, \{1, 2, 3\}\}.\end{aligned}$$

The first tuple has nonconstant entries  $S, P$ , so determines  $X_1$ . Intersecting the first two families leaves  $\{\{1, 2\}, \{1, 2, 3\}\} = \supseteq\{1, 2\}$ , the third family listed, whose only nonconstant content is  $S$ .

- (c) Recover  $X_1$  by taking the ordered pair  $(Y_{\{1,2\}}, Y_{\{1,3\}})$ . Recover  $Y_{\{1,2\}}$  from  $X_1$  by projecting onto its first coordinate.
- (d) Addresses  $\{1, 2\}$  and  $\{1, 3\}$  overlap but neither contains the other. Thus the contribution pattern need not form a nested tree.

The original model also satisfies the Markov condition. All nonconstant slots are mutually independent, so disjoint subcollections are independent, including after conditioning on the remaining common subcollection. Reconstruction follows from the indicated coordinate projections.

## 2.2 Two conditions on a latent model

LVM is not one of the two conditions below. It is part of the definition of a latent model, and every observable law admits one by construction: put a copy of each  $X_i$  in its singleton slot, or the entire tuple  $X$  in the top slot  $Y_I$ , and LVM holds either way. Satisfying it therefore says nothing about how information is shared. The two conditions below constrain these alternatives in different ways.

**Definition 2.3** (Reconstruction). A latent model has exact reconstruction if

$$H(Y_A \mid X_i) = 0 \quad \text{whenever } i \in A.$$

Equivalently,  $H(Y_{\supseteq A} \mid X_i) = 0$  whenever  $i \in A$ : every slot in that finite tuple can be recovered from  $X_i$ .

The equivalence in this definition uses exact zero errors. In approximate statements we will bound the entropy of the *whole tower*; separately small errors for its individual slots could add up.

A family  $\mathcal{F} \subseteq \mathcal{P}^+(I)$  is *upward closed* if  $A \in \mathcal{F}$  and  $A \subseteq C$  imply  $C \in \mathcal{F}$ . For example,  $\supseteq A$  and  $\mathcal{J}_B$  are upward closed. Taking a collection together with every slot above it prevents us from omitting higher variables on which it may depend.

**Definition 2.4** (Ordered Markov condition and defect). The ordered Markov condition is

$$I(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F} \cap \mathcal{G}}) = 0$$

for all upward-closed families  $\mathcal{F}, \mathcal{G} \subseteq \mathcal{P}^+(I)$ . Its quantitative defect is

$$\eta_Y = \max_{\mathcal{F}, \mathcal{G} \text{ upward closed}} I(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F} \cap \mathcal{G}}).$$

The conditioning family  $\mathcal{F} \cap \mathcal{G}$  is an intersection of *address families*, not a quantity inferred to be common by inspecting their values. The condition says that the latents at those shared addresses account for the dependence between the two collections. It does not require every latent to be independent of every other latent. In the two-observable case, its only potentially nonzero requirement is

$$I(Y_{\{1\}}; Y_{\{2\}} \mid Y_{\{1,2\}}) = 0.$$

This is a graphical-model condition for the inclusion order. The equivalent local version says that each slot is conditionally independent of all incomparable slots given its strict supersets.

**Definition 2.5** (Perfect condensation). A latent model is a perfect condensation if it has exact reconstruction and satisfies the ordered Markov condition.

This is equivalent to Eisenstat’s definition by optimal conditioned scores (Eisenstat, 2025, Theorem 5.10). Section 2.5 explains that connection. The overlapping-bit model in Example 2.1 is a perfect condensation.

**Exercise 2.2 (Two failed models; 20 minutes).** (★) Keep the observables of Example 2.1. Consider two alternative models.

- (a) Set  $Y_{\{i\}} = X_i$  for  $i = 1, 2, 3$  and every other slot constant. Check LVM and reconstruction. Take  $\mathcal{F} = \supseteq\{1\}$  and  $\mathcal{G} = \supseteq\{2\}$ . Calculate the conditional mutual information in Definition 2.4. Which information is duplicated across singleton slots?
- (b) Instead set  $Y_{\{1,2,3\}} = (S, P, Q)$  and every other slot constant. Check LVM and the Markov condition. Compute  $H(Y_{\{1,2,3\}} \mid X_1)$ . Which condition fails?
- (c) Explain why LVM, reconstruction, and the Markov condition have different roles.

### Solution to Exercise 2.2

- (a) In the singleton-copy model  $Y_{\{i\}} = X_i$ , the contributing family for  $i$  includes  $X_i$  itself. Thus LVM holds. Each nonconstant latent has just one contributing observation, from which it is recovered identically, so reconstruction holds too.

For  $\mathcal{F} = \supseteq\{1\}$  and  $\mathcal{G} = \supseteq\{2\}$ , all entries of  $Y_{\mathcal{F} \cap \mathcal{G}}$  are constant. The only nonconstant entries of the two larger tuples are  $X_1$  and  $X_2$ . Consequently

$$I(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F} \cap \mathcal{G}}) = I(X_1; X_2) = 1.$$

Both singleton slots determine  $S$ , while their common ancestor slots are constant. This violates the Markov condition. It is enough to exhibit this one pair of upward-closed families; finding the maximum defect is unnecessary.

- (b) With only  $Y_{\{1,2,3\}} = (S, P, Q)$  nonconstant, each observation is obtained by projection, so LVM holds. Every nonempty upward-closed family contains the top address. If both families are nonempty, their intersection contains that top slot and determines all latent variables; the conditional mutual information is zero. If either is empty it is zero as well. Thus the Markov condition holds.

But  $X_1 = (S, P)$  leaves the independent fair bit  $Q$  unknown, so

$$H(Y_{\{1,2,3\}} \mid X_1) = H(Q \mid S, P) = 1.$$

Reconstruction fails.

- (c) LVM requires enough latent information to determine each observation. Reconstruction excludes information placed at a shared address that an individual contributing observation cannot recover. The Markov condition excludes dependence between address collections not accounted for by their common slots. The singleton model satisfies the second requirement but fails the third; the top-slot model satisfies the third but fails the second.

**Check on the existence example.** Section 2.2.1 is a proved example, not an additional exercise. For a binary pair with full support, a top slot determined by each observable must be constant: writing it as  $f(X_1) = g(X_2)$ , each of the four pairs imposes an equality between the corresponding function values. With a constant top, LVM makes  $X_1$  a function of  $Y_{\{1\}}$  and  $X_2$  a function of  $Y_{\{2\}}$ . Thus data processing yields

$$I(X_1; X_2) \leq I(Y_{\{1\}}; Y_{\{2\}}) = \eta_Y.$$

The equality holds because, on the two-observable address poset, the only pair of incomparable nonempty upward-closed families consists of the two singleton towers. Every other Markov term is zero. For the noisy fair-bit pair,  $H(X_2) = 1$  and  $H(X_2 | X_1) = h_2(q)$ , proving the stated lower bound. It is attained by the singleton-copy model  $Y_{\{i\}} = X_i$ , with constant top. This model has both exact reconstruction directions but a positive Markov defect. Setting that defect to zero is impossible. The argument does not bound the trade-off when reconstruction is allowed to be inexact.

### 2.2.1 Existence is a separate question

The preceding examples test particular latent models. A stronger question is whether a given observable law admits *any* perfect condensation. Even a pair of correlated binary variables can fail to do so.

Let  $I = \{1, 2\}$ , let  $X_1$  be a fair bit, and set  $X_2 = X_1 \oplus N$ , where  $N$  is independent Bernoulli( $q$ ) noise and  $0 < q < 1/2$ . Here  $\oplus$  is addition modulo two, so  $a \oplus b$  is 0 when the two bits agree and 1 when they differ. The joint law is

| $P(x_1, x_2)$ | $x_2 = 0$       | $x_2 = 1$       |
|---------------|-----------------|-----------------|
| $x_1 = 0$     | $\frac{1-q}{2}$ | $\frac{q}{2}$   |
| $x_1 = 1$     | $\frac{q}{2}$   | $\frac{1-q}{2}$ |

Every one of the four outcomes has positive probability, because  $0 < q < 1/2$ ; and  $X_1, X_2$  are dependent, because the two rows differ. Suppose a perfect condensation existed, and consider its top slot  $Y_{\{1,2\}}$ .

With two observations there are only three addresses, so the whole model is three slots:

$$\begin{array}{c|ccc} \text{address} & \{1\} & \{1, 2\} & \{2\} \\ \text{slot} & Y_{\{1\}} & Y_{\{1,2\}} & Y_{\{2\}} \end{array}$$

LVM says  $X_1$  is determined by  $(Y_{\{1\}}, Y_{\{1,2\}})$  and  $X_2$  by  $(Y_{\{2\}}, Y_{\{1,2\}})$ . The top slot  $Y_{\{1,2\}}$  is the only one assigned to both observations, so it is the only place shared information can sit.

Reconstruction would give  $Y_{\{1,2\}} = f(X_1) = g(X_2)$  almost surely. Because all four observable pairs have positive probability, these equalities force

$$f(0) = g(0) = g(1) = f(1).$$

Thus  $Y_{\{1,2\}}$  is constant. The Markov condition would then require  $Y_{\{1\}}$  and  $Y_{\{2\}}$  to be independent. LVM makes each  $X_i$  a function of its singleton slot and the constant  $Y_{\{1,2\}}$ . Functions of independent variables are independent, contradicting the chosen observable law. No perfect condensation exists for this pair.

There is a quantitative obstruction if we retain *exact* reconstruction. The same argument makes  $Y_{\{1,2\}}$  constant. Data processing then gives

$$\eta_Y = I(Y_{\{1\}}; Y_{\{2\}}) \geq I(X_1; X_2) = 1 - h_2(q) > 0,$$

where  $h_2(q) = -q \log q - (1 - q) \log(1 - q)$  is the binary entropy function. Dependence between the observables must remain as a Markov defect when their separately recoverable top slot is constant.

That bound pins one end of a trade-off: exact reconstruction forces a Markov defect of at least  $1 - h_2(q)$ , and tolerating some reconstruction error buys part of it back. When no perfect condensation exists, which is the usual situation, the latent models of a given observable law trace out a Pareto frontier between reconstruction defect and Markov defect. The useful question is then whether some point on that frontier has both low *enough* for the correspondence bound of Section 2.3 to say something.

### Remark

One could instead ask for the model minimizing the conditioned score of Section 2.5. The score is defined per observation block, however, and there is not always a single model minimizing it everywhere.

**Exercise 2.3 (Collections, not individual slots; 15 minutes).** (★) For this exercise let  $I = \{1, 2\}$ , take independent fair bits  $C, R, T$ , and set  $X_1 = (C, R)$ ,  $X_2 = (C, T)$ . Compare

$$\begin{array}{lll} Y_{\{1,2\}} = C, & Y_{\{1\}} = R, & Y_{\{2\}} = T, \\ Z_{\{1,2\}} = C, & Z_{\{1\}} = (C, R), & Z_{\{2\}} = (C, T). \end{array}$$

- (a) Check that both models are perfect condensations.
- (b) Compute  $H(Z_{\{1\}} \mid Y_{\{1\}})$ .



- (c) Show that  $Y_{\supseteq\{1\}}$  and  $Z_{\supseteq\{1\}}$  determine one another. Explain why correspondence between these tuples is a different claim from correspondence between the singleton slots.

### Solution to Exercise 2.3

Here  $C, R, T$  are independent fair bits. In model  $Y$  the three slots are  $C, R, T$ ; in model  $Z$  they are  $C, (C, R), (C, T)$ .

- (a) Both contributing families determine their observations. All slots are recoverable from each observation to which they contribute. For example,  $C$  is the first coordinate of either  $X_1$  or  $X_2$ , and  $Z_{\{1\}} = X_1$ . For two observables the only nontrivial Markov requirement is independence of the two singleton slots given the top slot. For  $Y$  this is  $I(R; T | C) = 0$ . For  $Z$  it is

$$I((C, R); (C, T) | C) = I(R; T | C) = 0.$$

Both are therefore perfect condensations.

- (b)  $H(Z_{\{1\}} | Y_{\{1\}}) = H(C, R | R) = H(C) = 1$ .
- (c) The  $Y$  tower at  $\{1\}$  consists of  $(R, C)$ , and the  $Z$  tower consists of  $((C, R), C)$ . Each is a deterministic function of the other. A copy of the higher-address information can appear in a lower slot without changing the tower's total information. Thus the individual slots need not correspond even when the relevant towers do.

## 2.3 The correspondence theorem

Take two latent models of the same observable tuple  $X$ . To compare them, put  $(X, Y, Z)$  on one probability space while retaining both specified laws  $(X, Y)$  and  $(X, Z)$ . For finite variables this is always possible: for example, use  $p(x, y, z) = p(x)p(y | x)p(z | x)$ . The theorem holds for any such coupling.

**Theorem 2.6** (Correspondence). *For two latent models  $Y, Z$  of  $X$  and any nonempty  $A \subseteq I$ , let  $k = |A|$ . Then*

$$H(Y_{\supseteq A} | Z_{\supseteq A}) \leq \sum_{i \in A} H(Y_{\supseteq A} | X_i) + (k - 1)\eta_Z. \quad (2)$$

*If each term in the sum is at most  $\delta$  and  $\eta_Z \leq \eta$ , the bound is  $k\delta + (k - 1)\eta$ . If both budgets are  $\varepsilon$ , it is  $(2k - 1)\varepsilon$ . Interchanging  $Y, Z$  gives the reverse bound.*

**Corollary 2.7** (Exact correspondence). *If  $Y$  and  $Z$  are perfect condensations, their towers above every nonempty address  $A$  determine one another almost surely:*

$$H(Y_{\supseteq A} \mid Z_{\supseteq A}) = H(Z_{\supseteq A} \mid Y_{\supseteq A}) = 0.$$

The forward bound uses reconstruction in the sender  $Y$ , LVM in the receiver  $Z$ , and the Markov defect of  $Z$ . It does not use the Markov condition of  $Y$  until we reverse the argument. The result concerns corresponding collections. Exercise 2.3 shows why we cannot simply replace those collections by individual slots.

This proof is a specialization of Eisenstat’s approximate correspondence theorem (Eisenstat, 2025, Theorem 6.8), using the two-witness inequality (Eisenstat, 2025, Lemma 6.4). Its tower presentation follows the exposition listed in Section 6.1. The exercises below give the full argument without additional machinery.

### 2.3.1 Proof roadmap

The theorem has three steps.

1. If two witnesses each determine a target and are independent given their common part, that common part determines the target. Entropy gives a version with errors.
2. Each observation  $X_i$  is determined by its contributing  $Z$ -latents. Anything recoverable from  $X_i$  is therefore recoverable from that collection.
3. Intersect these contributing collections over  $i \in A$ . Exactly the addresses containing all of  $A$  survive. Each intersection costs at most one Markov error term.

The word “witness” below means a variable or tuple from which the target is recoverable.

**Lemma 2.8** (Two witnesses). *For any finite-valued variables  $U, M, P, Q$ ,*

$$H(U \mid M) \leq H(U \mid M, P) + H(U \mid M, Q) + I(P; Q \mid M). \quad (3)$$

**Exercise 2.4 (Prove the two-witness lemma; 20 minutes). (★)**

- (a) Prove Lemma 2.8 by expanding conditional entropies using the chain rule. Find an expression for the difference between the right and left sides that is a sum of nonnegative quantities.
- (b) Deduce the exact statement when the three terms on the right are zero.
- (c) Explain the three error terms. Which measure failures of reconstruction, and which measures dependence that remains after revealing  $M$ ?

**Hint:** Try expanding  $I(P; Q | M) - I(P; Q | U, M)$ , then rearranging with the chain rule, which is stated in Section 1.1. The desired difference contains a conditional mutual information and a conditional entropy.

### Solution to Exercise 2.4

For arbitrary finite  $U, M, P, Q$ , the exact identity is

$$\begin{aligned} H(U | M, P) + H(U | M, Q) + I(P; Q | M) - H(U | M) \\ = I(P; Q | U, M) + H(U | M, P, Q). \end{aligned}$$

For a direct verification, first subtract the two conditional mutual information terms using their entropy definitions:

$$\begin{aligned} I(P; Q | M) - I(P; Q | U, M) \\ = H(P | M) + H(Q | M) - H(P, Q | M) \\ \quad - H(P | U, M) - H(Q | U, M) + H(P, Q | U, M) \\ = I(U; P | M) + I(U; Q | M) - I(U; P, Q | M) \\ = H(U | M) - H(U | P, M) - H(U | Q, M) \\ \quad + H(U | P, Q, M). \end{aligned}$$

Rearrangement gives the displayed slack identity. Its right side is nonnegative, proving the lemma.

If  $H(U | M, P) = H(U | M, Q) = I(P; Q | M) = 0$ , the inequality and non-negativity imply  $H(U | M) = 0$ . Thus a target recoverable from each witness is recoverable from their designated common part when the witnesses are conditionally independent given that part.

The first two terms measure what each witness, together with  $M$ , still fails to determine about  $U$ . The third measures dependence between  $P, Q$  not accounted for by  $M$ . No independence assumption is needed for the inequality itself; exact independence is the case where the third error is zero.

**Exercise 2.5 (The dependence term is needed; 5 minutes).** Let  $U = P = Q$  be a fair bit and let  $M$  be constant. Evaluate every term in (3). What would go wrong if the mutual-information term were omitted? How does this example resemble the singleton-copy model in Exercise 2.2?

### Solution to Exercise 2.5

With  $U = P = Q$  a fair bit and  $M$  constant,

$$H(U | M) = 1, \quad H(U | M, P) = H(U | M, Q) = 0, \quad I(P; Q | M) = 1.$$

The inequality is the equality  $1 \leq 0 + 0 + 1$ . Without its mutual-information

term it would assert  $1 \leq 0$ . The two witnesses each determine the same bit while the designated common variable is constant, as in the singleton-copy model of Exercise 2.2.

**Exercise 2.6 (Transfer through an observable; 10 minutes).** (★) Let  $U$  be any finite-valued target and suppose  $X_i$  is determined by  $Z_{\supseteq\{i\}}$ . Prove

$$H(U \mid Z_{\supseteq\{i\}}) \leq H(U \mid X_i).$$

State the equality obtained by adding  $X_i$  to the conditioning tuple, then apply the conditioning inequality. Does this argument assume that  $U$  and  $Z_{\supseteq\{i\}}$  are conditionally independent given  $X_i$ ?

### Solution to Exercise 2.6

If  $X_i$  is a function of  $Z_{\supseteq\{i\}}$  almost surely, adding  $X_i$  to that conditioning tuple changes no information. Hence

$$H(U \mid Z_{\supseteq\{i\}}) = H(U \mid Z_{\supseteq\{i\}}, X_i) \leq H(U \mid X_i).$$

The inequality is conditioning monotonicity. It does not assume that  $U$  and the receiver tuple are conditionally independent given  $X_i$ . They can share additional information beyond  $X_i$ .

**Exercise 2.7 (Intersect the address families; 15 minutes).** (★)

- (a) For  $I = \{1, 2, 3\}$ , list  $\supseteq\{1\}$ ,  $\supseteq\{2\}$ , and  $\supseteq\{3\}$ . Compute  $\supseteq\{1\} \cap \supseteq\{2\}$  and then its intersection with  $\supseteq\{3\}$ .
- (b) Show that an intersection of upward-closed families is upward closed.
- (c) For general nonempty  $A \subseteq I$ , prove  $\bigcap_{i \in A} \supseteq\{i\} = \supseteq A$  by checking membership of an address  $C$  on both sides.
- (d) If  $\mathcal{F}, \mathcal{G}$  are upward closed, apply the two-witness lemma with  $M = Z_{\mathcal{F} \cap \mathcal{G}}$ ,  $P = Z_{\mathcal{F}}$ , and  $Q = Z_{\mathcal{G}}$ . Why can the extra conditioning on  $M$  be dropped from each reconstruction term? Write the resulting bound using  $\eta_Z$ .

### Solution to Exercise 2.7

- (a) The first two families are listed in Exercise 2.1, and

$$\supseteq\{3\} = \{\{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Thus  $\supseteq\{1\} \cap \supseteq\{2\} = \{\{1, 2\}, \{1, 2, 3\}\}$ , and intersecting with  $\supseteq\{3\}$  leaves

only  $\{1, 2, 3\}$ .

- (b) If  $C$  belongs to every family in an intersection and  $D \supseteq C$ , upward closure places  $D$  in each such family, hence in the intersection.
- (c) For an address  $C$ ,

$$C \in \bigcap_{i \in A} \supseteq \{i\} \iff (\forall i \in A) i \in C \iff A \subseteq C \iff C \in \supseteq A.$$

- (d) Put  $M = Z_{\mathcal{F} \cap \mathcal{G}}$ ,  $P = Z_{\mathcal{F}}$ ,  $Q = Z_{\mathcal{G}}$ . The tuple  $M$  is a subtuple of both  $P$  and  $Q$ . Thus  $H(U \mid M, P) = H(U \mid P)$  and similarly for  $Q$ . By the definition of  $\eta_Z$ , the lemma gives

$$H(U \mid Z_{\mathcal{F} \cap \mathcal{G}}) \leq H(U \mid Z_{\mathcal{F}}) + H(U \mid Z_{\mathcal{G}}) + \eta_Z.$$

Upward closure is needed to invoke the bound  $\eta_Z$ .

**Exercise 2.8 (Complete the correspondence proof; 25 minutes). (★)**

Fix  $A = \{i_1, \dots, i_k\}$ , set  $\mathcal{G}_j = \bigcap_{\ell=1}^j \supseteq \{i_\ell\}$ .

- (a) Use Exercise 2.6 to prove the bound at  $j = 1$ .
- (b) Use Exercise 2.7 to show, for  $j \geq 2$ ,

$$H(Y_{\supseteq A} \mid Z_{\mathcal{G}_j}) \leq H(Y_{\supseteq A} \mid Z_{\mathcal{G}_{j-1}}) + H(Y_{\supseteq A} \mid Z_{\supseteq \{i_j\}}) + \eta_Z.$$

- (c) Prove by induction that

$$H(Y_{\supseteq A} \mid Z_{\mathcal{G}_j}) \leq \sum_{\ell=1}^j H(Y_{\supseteq A} \mid X_{i_\ell}) + (j-1)\eta_Z.$$

Evaluate this at  $j = k$  to prove Theorem 2.6. Check the case  $k = 1$  explicitly.

- (d) Deduce Corollary 2.7, including the reverse direction. Name the assumption used at each step of the proof.
- (e) Suppose instead that  $H(Y_C \mid X_i) \leq \varepsilon$  separately for all relevant slots  $C$ . Why does this not by itself supply the tower error budget  $H(Y_{\supseteq A} \mid X_i) \leq \varepsilon$ ?

## Solution to Exercise 2.8

Fix a nonempty  $A = \{i_1, \dots, i_k\}$ , put  $\mathcal{G}_j = \bigcap_{\ell=1}^j \supseteq \{i_\ell\}$ . Every  $\supseteq \{i\}$  is upward closed, as is every  $\mathcal{G}_j$  by Exercise 2.7. We prove

$$H(Y_{\supseteq A} \mid Z_{\mathcal{G}_j}) \leq \sum_{\ell=1}^j H(Y_{\supseteq A} \mid X_{i_\ell}) + (j-1)\eta_Z \quad (4)$$

for  $1 \leq j \leq k$ .

For  $j = 1$ ,  $\mathcal{G}_1 = \supseteq \{i_1\}$ , and Exercise 2.6 gives  $H(Y_{\supseteq A} \mid Z_{\mathcal{G}_1}) \leq H(Y_{\supseteq A} \mid X_{i_1})$ . This is the claimed bound with zero intersection terms.

Suppose (4) holds for  $j-1$ , where  $2 \leq j \leq k$ . Apply Exercise 2.7 to the upward-closed families  $\mathcal{G}_{j-1}$  and  $\supseteq \{i_j\}$ . Their intersection is  $\mathcal{G}_j$ , so

$$\begin{aligned} H(Y_{\supseteq A} \mid Z_{\mathcal{G}_j}) &\leq H(Y_{\supseteq A} \mid Z_{\mathcal{G}_{j-1}}) + H(Y_{\supseteq A} \mid Z_{\supseteq \{i_j\}}) + \eta_Z \\ &\leq \sum_{\ell=1}^{j-1} H(Y_{\supseteq A} \mid X_{i_\ell}) + (j-2)\eta_Z + H(Y_{\supseteq A} \mid X_{i_j}) + \eta_Z \\ &= \sum_{\ell=1}^j H(Y_{\supseteq A} \mid X_{i_\ell}) + (j-1)\eta_Z. \end{aligned}$$

The second line uses the induction hypothesis and Exercise 2.6. This completes the induction. At  $j = k$ , Exercise 2.7 gives  $\mathcal{G}_k = \supseteq A$ , hence

$$H(Y_{\supseteq A} \mid Z_{\supseteq A}) \leq \sum_{i \in A} H(Y_{\supseteq A} \mid X_i) + (|A| - 1)\eta_Z.$$

For  $|A| = 1$  this is exactly the observable-transfer inequality, so no intersection or Markov error is required. There are  $k$  reconstruction errors and  $k-1$  Markov defects. Bounding each reconstruction error by  $\delta$  and each Markov defect by  $\eta$  gives  $k\delta + (k-1)\eta$ , or  $(2k-1)\varepsilon$  for a common budget.

If  $Y, Z$  are perfect condensations, each reconstruction error is zero: for every  $C \supseteq A$  and  $i \in A$ , reconstruction says  $Y_C$  is a function of  $X_i$ . The finite tuple of such variables is consequently a function of  $X_i$ . Also  $\eta_Z = 0$ . The inequality gives zero conditional entropy in the forward direction. Interchanging  $Y, Z$ , using reconstruction in  $Z$  and  $\eta_Y = 0$ , proves the reverse direction. Zero conditional entropy is almost-sure deterministic recoverability in our finite setting.

The proof uses LVM in  $Z$  for Exercise 2.6, the sender's reconstruction to bound those errors, and the receiver's Markov condition at intersections. It needs no cross-model independence, and its finite coupling preserves all marginal conditions used.

Finally, approximate slotwise errors do not directly give the same error for a tuple. For example, if two independent fair bits  $R, T$  remain unknown given a constant observation, each conditional entropy is one bit but the tuple's is two

bits. Conditional subadditivity gives a sum of the slot budgets, not a single budget. The theorem states its hypothesis in terms of tower errors to avoid this ambiguity.

## 2.4 Optional: several observations as one witness

A shared parameter need not be recoverable from a single observation. For example, take  $n \geq 2$ , let  $C$  be a fair bit, and set  $X_i = C \oplus N_i$ , where the  $N_i$  are independent Bernoulli( $p$ ) noises with  $0 < p < 1/2$ , independent of  $C$ . The model  $Y_I = C$ ,  $Y_{\{i\}} = X_i$  satisfies LVM and the Markov condition exactly, but  $H(C \mid X_i) = h_2(p) > 0$ , where  $h_2$  is the binary entropy function of Section 2.2.1. Larger observation blocks can reveal more about  $C$ . To use them in the proof, replace  $\supseteq\{i\}$  by  $\mathcal{J}_B$ .

### Exercise 2.9 (Block witnesses; optional, 25 minutes).

- (a) Let  $I = \{1, 2, 3\}$  and suppose a target  $U$  has conditional entropy at most  $\delta$  given each of  $X_{\{1,2\}}$ ,  $X_{\{1,3\}}$ , and  $X_{\{2,3\}}$ . Which addresses survive the intersection  $\mathcal{J}_{\{1,2\}} \cap \mathcal{J}_{\{1,3\}} \cap \mathcal{J}_{\{2,3\}}$ ? Prove a bound on the entropy of  $U$  given the surviving  $Z$ -slots.
- (b) More generally, let  $\mathcal{W}$  be a nonempty finite family of nonempty observation blocks. Adapt Exercise 2.8 to show

$$H(U \mid Z_{\mathcal{R}}) \leq \sum_{B \in \mathcal{W}} H(U \mid X_B) + (|\mathcal{W}| - 1)\eta_Z, \quad \mathcal{R} = \bigcap_{B \in \mathcal{W}} \mathcal{J}_B.$$

- (c) Fix an integer  $0 \leq r < |A|$  and take every  $(r+1)$ -element subset of  $A$  as a witness. Show that  $C \in \mathcal{R}$  exactly when  $|A \setminus C| \leq r$ . What does this give for  $r = 0$ ? Why is the receiver generally larger for  $r > 0$ ?

### Solution to Exercise 2.9

- (a) A nonempty address  $C \subseteq \{1, 2, 3\}$  meets each of  $\{1, 2\}$ ,  $\{1, 3\}$ ,  $\{2, 3\}$  precisely when  $|C| \geq 2$ . A singleton misses the complementary pair; every address with at least two elements meets every pair. Thus the surviving tuple consists of the three pair slots and the top slot. Applying the two-witness lemma twice, after observable transfer for each block, gives

$$H(U \mid Z_{\{C: |C| \geq 2\}}) \leq 3\delta + 2\eta_Z.$$

It does not isolate the top slot alone.

- (b) For each block  $B$ , the collection  $Z_{\mathcal{J}_B}$  determines  $X_B$ : for every  $i \in B$  it contains  $Z_{\supseteq\{i\}}$ . Exercise 2.6 therefore extends to  $H(U \mid Z_{\mathcal{J}_B}) \leq H(U \mid X_B)$ . Each  $\mathcal{J}_B$  is upward closed. Enumerate  $\mathcal{W} = \{B_1, \dots, B_N\}$  and intersect

the first  $j$  contributing families. Exactly the induction in Exercise 2.8, now with these families as leaves, gives

$$H(U \mid Z_{\mathcal{R}}) \leq \sum_{B \in \mathcal{W}} H(U \mid X_B) + (N - 1)\eta_Z, \quad \mathcal{R} = \bigcap_{B \in \mathcal{W}} \mathcal{J}_B.$$

For  $N = 1$  this is just the transfer inequality. We required a nonempty family so the induction and the intersection count apply.

- (c) An address  $C$  fails to meet some  $(r + 1)$ -subset of  $A$  exactly when  $A \setminus C$  contains at least  $r + 1$  elements. Thus it meets every such subset exactly when  $|A \setminus C| \leq r$ . For  $r = 0$  this says  $A \subseteq C$ , recovering the tower. For  $r > 0$  an address can survive even if it misses some elements of  $A$ ; the receiver is a larger tuple. If every witness error is at most  $\delta$ , write  $N = \binom{|A|}{r+1}$  to obtain the explicit bound  $N\delta + (N - 1)\eta_Z$ . The restriction  $r < |A|$  ensures witnesses exist.

Eisenstat’s later almost-perfect-condensation formulation (Eisenstat, 2025) bounds the tower reconstruction error whenever  $|A \cap B| > r$ , the Markov defect, and the number of active slots. The threshold allows small slots to account for local noise without requiring every such slot to be recovered from every relevant block. It also changes which receiver the proof isolates, as Exercise 2.9 shows. The exact-tower bound with singleton witnesses does not follow from an arbitrary positive threshold  $r$ .

## 2.5 Optional: the conditioned score

The conditions also have a description-length interpretation. For an observation block  $B$ , compare

$$\sigma(B) = \sum_{A \in \mathcal{J}_B} H(Y_A), \quad \chi(B) = \sum_{A \in \mathcal{J}_B} H(Y_A \mid Y_{\supseteq A \setminus \{A\}}).$$

The simple score  $\sigma$  charges each active slot separately. The conditioned score  $\chi$  charges its additional information given strict supersets. Those supersets are also active for  $B$ . Encoding in reverse inclusion order makes this conditioning available. These are ideal expected description-length quantities; they do not measure computational retrieval time.

Both scores are at least  $H(X_B)$ . More precisely, put  $V_B = \chi(B) - H(Y_{\mathcal{J}_B})$ . Then

$$\chi(B) - H(X_B) = V_B + H(Y_{\mathcal{J}_B} \mid X_B), \quad V_B \geq 0. \quad (5)$$

The second term charges latent information not recoverable from  $X_B$ . The first charges dependence not accounted for by the inclusion hierarchy. It vanishes under the ordered Markov condition.

For a graphical-model expression, define

$$q_B(y) = \prod_{A \in \mathcal{J}_B} p(y_A \mid y_{\supseteq A \setminus \{A\}}).$$



This is a normalized distribution: sample larger addresses before smaller ones. Conditional kernels on probability-zero parent configurations can be chosen arbitrarily. Direct expansion gives  $V_B = D_{\text{KL}}(p_{Y_{\mathcal{J}_B}} \| q_B)$ . Thus (5) separates two nonnegative defects.

**Exercise 2.10 (The two charges; optional, 25 minutes).**

- (a) Order  $\mathcal{J}_B$  so strict supersets appear first. Use the chain rule and conditioning inequality to prove  $\chi(B) \geq H(Y_{\mathcal{J}_B})$ .
- (b) Use LVM to prove  $H(Y_{\mathcal{J}_B}) - H(X_B) = H(Y_{\mathcal{J}_B} | X_B)$ , hence (5).
- (c) For both models in Exercise 2.3, calculate  $\sigma(\{1, 2\})$  and  $\chi(\{1, 2\})$ . Why is copying the top variable into a lower slot free under one score but not the other?
- (d) If  $A \cap B \neq \emptyset$ , show that  $H(Y_{\supseteq A} | X_B) \leq \chi(B) - H(X_B)$ .

**Solution to Exercise 2.10**

- (a) List  $\mathcal{J}_B$  as  $A_1, \dots, A_q$  with strict supersets appearing before subsets. Such an order exists because strict inclusion has no cycles. Every strict superset of an active address also meets  $B$ , so is in this list. Therefore

$$H(Y_{A_t} | Y_{\supseteq A_t \setminus \{A_t\}}) \geq H(Y_{A_t} | Y_{A_1}, \dots, Y_{A_{t-1}}).$$

Summing and applying the chain rule gives  $\chi(B) \geq H(Y_{\mathcal{J}_B})$ , hence  $V_B \geq 0$ .

- (b) Since  $Y_{\mathcal{J}_B}$  determines  $X_B$ , the joint entropy can be expanded in two ways:

$$H(X_B, Y_{\mathcal{J}_B}) = H(Y_{\mathcal{J}_B}) = H(X_B) + H(Y_{\mathcal{J}_B} | X_B).$$

Subtracting  $H(X_B)$  and adding  $V_B$  proves

$$\chi(B) - H(X_B) = V_B + H(Y_{\mathcal{J}_B} | X_B).$$

- (c) In Exercise 2.3, the independent model  $Y$  has simple score  $1 + 1 + 1 = 3$  and conditioned score  $1 + 1 + 1 = 3$ . In model  $Z$ , the simple score is  $1 + 2 + 2 = 5$ , while the conditioned score is

$$H(C) + H(C, R | C) + H(C, T | C) = 1 + 1 + 1 = 3.$$

Both models determine  $X_1, X_2$ , whose joint entropy is three. The conditioned score charges only the extra information below  $C$ ; its copies add zero conditional entropy. The simple score charges those copies again.

- (d) If  $A \cap B \neq \emptyset$ , every  $C \supseteq A$  meets  $B$ . Thus  $Y_{\supseteq A}$  is a subtuple of  $Y_{\mathcal{J}_B}$  and

$$H(Y_{\supseteq A} | X_B) \leq H(Y_{\mathcal{J}_B} | X_B) \leq \chi(B) - H(X_B),$$

where the last inequality uses the nonnegative score decomposition.

**Additional check: equivalence of the perfect definitions.** This argument is available for students who want to connect the two definitions completely; it is not an extra required exercise.

Let  $V_I = \chi(I) - H(Y_{\mathcal{P}^+(I)})$ . For upward-closed  $\mathcal{F}, \mathcal{G}$ , partition  $\mathcal{P}^+(I)$  into

$$\mathcal{C} = \mathcal{F} \cap \mathcal{G}, \quad \mathcal{P} = \mathcal{F} \setminus \mathcal{G}, \quad \mathcal{Q} = \mathcal{G} \setminus \mathcal{F}, \quad \mathcal{R} = \mathcal{P}^+(I) \setminus (\mathcal{F} \cup \mathcal{G}).$$

Order these blocks  $\mathcal{C}, \mathcal{P}, \mathcal{Q}, \mathcal{R}$ , internally by reverse inclusion. Upward closure ensures this order is valid and no slot in  $\mathcal{Q}$  has a strict superset in  $\mathcal{P}$ , or conversely. Conditioning inequalities and chain rules within each block imply

$$\begin{aligned} \chi(I) &\geq H(Y_{\mathcal{C}}) + H(Y_{\mathcal{P}} \mid Y_{\mathcal{C}}) + H(Y_{\mathcal{Q}} \mid Y_{\mathcal{C}}) + H(Y_{\mathcal{R}} \mid Y_{\mathcal{C}}, Y_{\mathcal{P}}, Y_{\mathcal{Q}}) \\ &= H(Y_{\mathcal{P}^+(I)}) + I(Y_{\mathcal{P}}; Y_{\mathcal{Q}} \mid Y_{\mathcal{C}}). \end{aligned}$$

The latter mutual information equals  $I(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F} \cap \mathcal{G}})$ . Hence  $\eta_Y \leq V_I$ .

If all score gaps vanish, the decomposition forces  $V_I = 0$ , giving the Markov condition. The singleton gap at  $B = \{i\}$  forces the contributing tuple, and therefore each slot at an address containing  $i$ , to be determined by  $X_i$ . This gives reconstruction.

Conversely, suppose reconstruction and the Markov condition hold. For every  $B$ , each active slot  $Y_A$  is determined by some  $X_i$  with  $i \in A \cap B$ , and hence by  $X_B$ . Therefore  $H(Y_{\mathcal{J}_B} \mid X_B) = 0$ . In a reverse-inclusion order, the earlier slots consist of all strict supersets of the current slot and some incomparable slots. The global Markov condition implies that conditioning on these extra earlier slots does not reduce its entropy beyond conditioning on strict supersets: apply the condition to the upward cone of the current address and the upward-closed family of earlier addresses. Their intersection is the strict-superset family. Thus the score's chain-rule inequality is an equality,  $V_B = 0$ . The decomposition gives  $\chi(B) = H(X_B)$  for all  $B$ .

Eisenstat defines perfect condensation by  $\chi(B) = H(X_B)$  for every  $B$ . The equivalence with the characterization above is his Theorem 5.10. The decomposition above also explains why small score gaps imply strong reconstruction conditions. The converse for positive-threshold almost-perfect condensation requires additional reconstruction coverage: some active slots may never meet that threshold. Do not identify the two approximate criteria without checking these quantifiers.

## 2.6 What the result says, and connections

The theorem compares two joint laws extending the same observable law. Under the exact conditions, the two latent towers at each address determine one another almost surely; under the quantitative hypotheses, their conditional entropies satisfy the stated bounds. The theorem does not select the observable family or give a learning algorithm. A perfect condensation need not exist, as the full-support example in Section 2.2.1

shows.

### 2.6.1 Connections to established questions

In statistics, a statistic is sufficient for a parameter when it retains the data's information relevant to that parameter. Predictive sufficiency instead concerns retaining information about a target to be predicted. Condensation studies the organization of information across an indexed family of observations, with explicit contribution addresses and a correspondence result. It shares questions about information retention with sufficiency, but these are not identical definitions or claims.

Graphical models supply the language of conditional independence and factorization used by the ordered Markov condition. The graph here is fixed by inclusion of addresses; its edges do not by themselves have a causal meaning. Common-information questions also ask what two variables share. Our correspondence argument applies across a family of overlapping address collections, using the same entropy calculus rather than a new information measure. Computational mechanics studies predictive states obtained from histories with the same conditional future law. Condensation's formulation is static: time, learning, and a prescribed past/future split are not required. None of these comparisons implies that earlier frameworks cannot study structured or modular organizations of information.

### 2.6.2 Limits of information-theoretic correspondence

If  $V = f(U)$  for a bijection  $f$ , then  $H(U | V) = H(V | U) = 0$ . This remains true when describing or discovering  $f$  is expensive. The theorem does not provide an efficient translation algorithm or account for the description length of the probability law.

Small conditional entropy is an average guarantee. If  $E$  is Bernoulli( $p$ ),  $D$  is an independent fair bit,  $U = (E, ED)$ , and  $V = E$ , then  $H(U | V) = p$ . On the rare event  $E = 1$ , the missing bit is still completely unknown. Conversely, a rare event with an enormous number of outcomes can dominate an entropy. A safety application needs to say which failures matter, not just report an average information budget.

### 2.6.3 Discussion prompts

1. Choose a comparison from Section 4.2. State one mathematical similarity and one difference in what is being characterized. If it relates to one of the condensation conditions, say how.
2. Corresponding latent collections in two models determine one another with zero conditional entropy. What does this establish, and what else might be needed to find and use the translation between them?
3. Where might a correspondence like this matter for AI safety? Trace one chain from the theorem to something you would want, and say where it breaks: which steps does the theorem not establish? Section 4.4 sets out one such chain.

### 3 A hierarchical clustering example

#### 3.1 Thirty-six observations

There are two groups, each containing two clusters, each with nine observations. Write  $X_{gcj}$  for the  $j$ th two-dimensional point in cluster  $c$  of group  $g$ . The labels  $(g, c, j)$  and memberships are fixed. We are comparing models of this labeled family, not inferring an unknown partition of an unlabeled cloud.

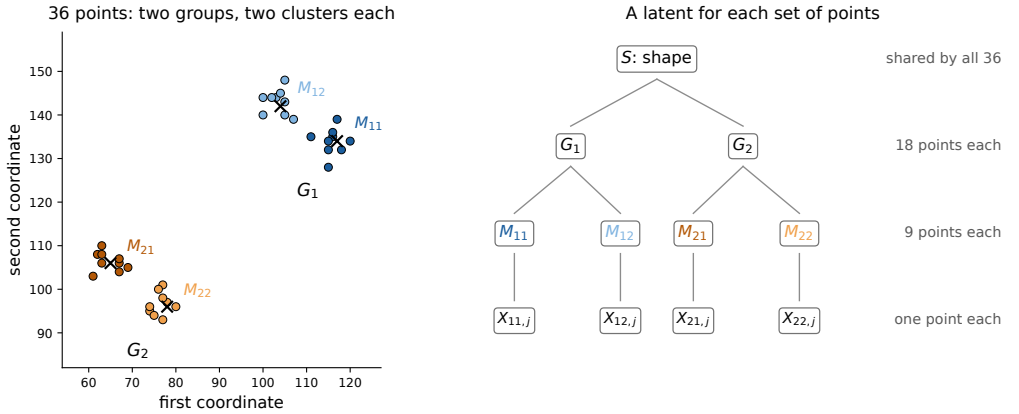


Figure 1: One draw from the finite model. The left panel shows the 36 points; the right panel shows which set of observations receives each parameter, and is not a graph inferred from the points. The global shape  $S$  is available to every observation.

The parameters are a global shape  $S$ , group centers  $G_1, G_2$ , and cluster centers  $M_{gc}$ . Given these parameters, the 36 observations are independent. Marginalizing over shared parameters generally makes observations dependent. The plotted cloud is one realization; the random variables and their entropies refer to the joint law across realizations.

#### 3.2 An explicit finite law

Let  $D = \{0, \dots, 200\}^2$ . Draw  $S$  uniformly from three shape labels, and draw  $G_1, G_2$  independently and uniformly from  $\{60, \dots, 140\}^2$ , independently of  $S$ . Conditional on  $G_g$ , draw its two centers independently:

$$P(M_{gc} = m \mid G_g = a) \propto \exp\left(-\frac{\|m - a\|^2}{128}\right), \quad m \in D.$$

Given  $S$  and all centers, draw the observations independently, using

$$P(X_{gcj} = x \mid M_{gc} = m, S = s) \propto \exp\left(-\frac{1}{2}(x - m)^\top Q_s^{-1}(x - m)\right), \quad x \in D,$$

where

$$Q_{\text{round}} = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}, \quad Q_{\text{right}} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}, \quad Q_{\text{left}} = \begin{pmatrix} 25 & -20 \\ -20 & 25 \end{pmatrix}.$$

Each proportionality means normalization by the sum over  $D$ . These are probability mass functions on a finite grid. The matrices control shape; they are not the exact covariances after discretization and boundary truncation. Every grid point has positive probability. Cluster centers tend to lie near their group center, but the four components need not be visually separated in every realization.

### 3.3 A structured latent model

Let  $I$  contain all 36 indices. Let  $A_g$  be the 18 indices of group  $g$ , and  $A_{gc}$  the nine indices of cluster  $(g, c)$ . Assign

$$Y_I = S, \quad Y_{A_g} = G_g, \quad Y_{A_{gc}} = M_{gc}, \quad Y_{\{i\}} = X_i,$$

and make every other slot constant. The joint law factorizes in the address inclusion order, so its ordered Markov defect is zero.

The singleton slot supplies the sampled point itself. Thus  $H(X_i | Y_{\{A: i \in A\}}) = 0$  exactly. The cluster center and shape alone specify a conditional distribution for the point; they do not determine its realized coordinates. The residual conditional entropy is  $H(X_i | M_{gc}, S)$ .

Reconstruction is different. All the kernels have full support, so any finite observation block leaves several parameter configurations with positive posterior probability. For example,  $H(M_{gc} | X_i) > 0$ . The model's joint law satisfies the Markov condition exactly, but it is not a perfect condensation.

For a nonempty block  $B$ , the general score-gap identity (5) is

$$\chi(B) - H(X_B) = [\chi(B) - H(Y_{\mathcal{J}_B})] + H(Y_{\mathcal{J}_B} | X_B).$$

Here the bracketed Markov term is zero. The gap is the latent information remaining uncertain after observing the queried block.

### 3.4 Three organizations of the same observable law

For a nonempty queried *observation block*  $B$ , the conditioned score sums the charges of all slots whose addresses meet  $B$ .

**One global tuple.** Put  $Z_I = X_I$  and make every other slot constant. The global slot determines each observation. For every nonempty  $B$ ,

$$\chi_{\text{global}}(B) = H(X_I), \quad \chi_{\text{global}}(B) - H(X_B) = H(X_I | X_B).$$

This attains the entropy bound when querying the whole tuple. Even a singleton query incurs the entire tuple's entropy in this score.

**An ordered chain.** Choose an ordering  $X_1, \dots, X_{36}$ . Set  $Z_{\{j,j+1,\dots,36\}} = X_j$  and make other slots constant. The contributing slots for  $X_i$  contain  $X_1, \dots, X_i$ . The charge for slot  $j$  is  $H(X_j \mid X_1, \dots, X_{j-1})$ , hence

$$\chi_{\text{chain}}(B) = H(X_1, \dots, X_{\max B}).$$

This attains the entropy bound for prefix queries. Querying only  $X_{36}$  costs  $H(X_I)$ . These statements concern this particular address assignment, not every autoregressive architecture.

**The structured model.** The score charges the shape once, the group and cluster parameters needed by  $B$ , and the residual uncertainty of each requested observation:

$$\begin{aligned} \chi_{\text{structured}}(B) = & H(S) + \sum_{g: B \cap A_g \neq \emptyset} H(G_g) \\ & + \sum_{g,c: B \cap A_{gc} \neq \emptyset} H(M_{gc} \mid G_g) + \sum_{i \in B} H(X_i \mid M_{g(i)c(i)}, S). \end{aligned}$$

This expression avoids charging for unrelated observations. It need not be smaller than the other scores for every block. In particular, for  $B = I$  it exceeds  $H(X_I)$  by the remaining parameter uncertainty, whereas both baseline constructions attain  $H(X_I)$ . Fitting the observable law does not by itself settle which organization is preferable across queries.

### 3.5 Why blocks help recover parameters

For nonempty  $B$  inside a fixed cluster  $(g, c)$ , set  $\Theta = (S, G_g, M_{gc})$ . The nonconstant contributing tuple is  $(\Theta, X_B)$ , so

$$\chi_{\text{structured}}(B) - H(X_B) = H(\Theta \mid X_B).$$

For nonempty  $B \subseteq B' \subseteq A_{gc}$ , the reduction in this gap is

$$H(\Theta \mid X_B) - H(\Theta \mid X_{B'}) = I(\Theta; X_{B' \setminus B} \mid X_B) \geq 0.$$

More points can inform the shared parameters. Exact recovery still fails for finite blocks. Even knowing one cluster center need not determine its parent group center. When  $B$  extends to another cluster, its contributing parameter family changes, so the displayed monotonicity argument no longer applies as written. This example motivates block witnesses and nonzero recovery errors; it does not establish a chosen uniform error bound.

**Source.** The motivating hierarchy is from Gillen and Chiang, *A summary of Condensation and its relation to Natural Latents* (Gillen and Chiang, 2026), §1.1.1. The finite law, illustration, and exact score comparisons above are this course’s adaptation.

## 4 Condensation in context

### 4.1 What is being characterized?

Fix jointly distributed observables  $X_1, \dots, X_n$ . We organize information into latents  $Y_A$ , indexed by nonempty subsets  $A$  of the observable indices. The address specifies which observations the latent is assigned to. Each  $X_i$  is determined by its contributing collection ( $Y_A : i \in A$ ).

LVM alone leaves considerable freedom. The additional reconstruction and Markov conditions constrain where shared information can reside. The correspondence theorem compares two models satisfying these conditions: their collections above the same address determine one another, exactly or with a controlled conditional-entropy error.

This provides a conditional sense of objectivity. Once the observables and the conditions are fixed, agreement in information content need not come from a shared vocabulary. It follows from each model's relationship to the observations. The theorem does not choose the observables. Changing the questions changes the problem, and may change the information that must correspond.

Nor is each latent automatically a human concept. A model includes residual information needed to determine observations. The interpretation of its parts as concepts is an additional connection between the mathematics and the uses of a latent model.

### 4.2 Connections to prior work

#### 4.2.1 Sufficient statistics

For a fixed joint law of  $(X, V)$ , a statistic  $T = f(X)$  is sufficient for predicting  $V$  when

$$I(X; V \mid T) = 0.$$

Knowing  $X$  then gives no predictive information about  $V$  beyond  $T$ . For finite alphabets, grouping values  $x$  with the same conditional distribution  $P(V \mid X = x)$  gives a minimal such statistic, on the support of  $X$ . This differs from parametric sufficiency, where the conditional law of the sample given a statistic must be independent of the unknown parameter. Both formulations ask which distinctions must be retained for an inference task.

Predictive sufficiency does not say that  $T$  determines the realized value of  $V$ . Condensation's LVM condition does demand determination, using the whole contributing collection, including residual variables. A sufficient statistic can itself be a vector or another structured object.

A sufficient statistic is by construction a function of  $X$ . A latent need not be, because reconstruction is asked for only up to a defect: that leaves room for latents which are parameters of a law rather than functions of the observations. The additional question here is how to organize a family of latent parts across many specified observables. See [Fisher \(1922\)](#) and Shalizi's account of predictive sufficiency ([Shalizi, 2019](#)).

### 4.2.2 Common information

Two older questions distinguish requirements that appear together in the perfect case. The maximal deterministic common part of  $X_1, X_2$  is a variable  $K$  recoverable from either observation, through which every other common function factors. Its entropy is the Gács–Körner common information. Wyner’s common information instead minimizes  $I(X_1, X_2; W)$  over extensions satisfying  $X_1 \perp X_2 \mid W$ . Such a  $W$  need not be separately recoverable from the observations.

If  $W$  is both separately recoverable and screens off dependence, then

$$I(X_1; X_2) = H(W) + I(X_1; X_2 \mid W) = H(W).$$

Moreover, every common function  $K$  is determined by  $W$ : the two-witness lemma, with witnesses  $X_1, X_2$  and common part  $W$ , gives  $H(K \mid W) = 0$ . Thus this special case is closely connected to common information in the sense of Gács and Körner.

It is restrictive. If  $X$  is a fair bit and  $V = X \oplus N$  for independent  $N \sim \text{Bernoulli}(p)$ ,  $0 < p < 1/2$ , all four pairs have positive probability. Any common function must therefore be constant, although  $I(X; V) = 1 - h_2(p) > 0$ , where  $h_2$  is the binary entropy function of Section 2.2.1. Recoverable common information and mutual information differ.

Common/private source coding already studies structured organization of information. Condensation’s particular focus is a family of subset-addressed latents and correspondence under conditions on that family. See Gács and Körner (1973) and Liu, Xu and Chen’s extension of Wyner’s common information (Liu et al., 2010).

### 4.2.3 Computational mechanics

Computational mechanics groups past histories according to their conditional distributions over futures. Its causal states have predictive optimality, minimality, and uniqueness properties. This is an important precedent for characterizing such a construction through informational requirements.

The present setup starts from a family of jointly distributed questions rather than a distinguished past/future split. It characterizes latent collections associated with different observable subsets. See Shalizi and Crutchfield (2001).

### 4.2.4 Graphical models and the information bottleneck

The ordered Markov condition in condensation is a graphical-model condition. A subset-ordered directed graph places larger addresses above smaller ones. Under the corresponding factorization, the latent joint law is

$$P(y) = \prod_{A \neq \emptyset} P(y_A \mid y_{\supseteq A}).$$

A factorization alone does not ensure that observations recover the latents, or that different latent models correspond. Those are additional requirements. Graphical lan-



guage also does not make the result a causal-identification theorem. Eisenstat discusses this connection explicitly in Definition 5.8 and Example 6.3 of the condensation paper (Eisenstat, 2025).

The information bottleneck compresses an input  $X$  into  $T$  while retaining information about a specified relevance variable  $V$ . A standard objective is

$$I(X;T) - \beta I(T;V), \quad T \perp V \mid X.$$

It provides an established distribution-based approach to useful summaries of the data. Our condensation problem instead involves many addressed latents, LVM, and conditions across observable sets. An optimum of a generic bottleneck objective does not automatically satisfy these conditions. See Tishby et al. (1999).

### 4.3 What information theory leaves out

#### 4.3.1 An encoding and access to its inverse

Let  $X$  be uniform on  $\{1, \dots, N\}$  and let  $Y = \pi(X)$  for a fixed permutation  $\pi$ . Then

$$H(X \mid Y) = H(Y \mid X) = 0.$$

This is exact correspondence whether  $\pi$  is the identity or a large unstructured lookup table. The entropies do not charge for describing the table, learning it, storing it, or using it. For a finite table an inverse exists and is computable; that does not make it available to a particular learner.

For a concrete learning distinction, draw  $\pi$  uniformly from all permutations and reveal  $m < N$  distinct input–output pairs. Conditional on those pairs, the image of a previously unseen input is uniform over the  $N - m$  unused outputs. The best success probability without further information is  $1/(N - m)$ . For every fixed permutation the correspondence entropy is zero, yet a learner with few examples cannot predict an unseen pairing. These are different probability models: one fixes the permutation; the other describes uncertainty about it.

A separate question concerns computation even when a transformation has a short known description. Entropic bounds alone give no running-time bound for an inverse. Establishing computational difficulty requires an additional argument or assumption; it does not follow from this permutation example.

#### 4.3.2 Rare events and average error

Let  $E \sim \text{Bernoulli}(p)$ . Define  $R = 0$  when  $E = 0$  and, when  $E = 1$ , let  $R$  be uniform over  $M$  other values. Since  $R$  reveals  $E$ ,

$$H(R) = h_2(p) + p \log_2 M.$$

For  $p = 10^{-6}$  and  $M = 2^{10^6}$ , the rare branch contributes one full bit. An independent sample of size  $m$  misses that branch with probability  $(1 - p)^m$ . Large informational contributions can be absent from a small dataset.

Conversely, suppose a variable  $T$  records  $E$ , and records the value of a fair bit  $B$ , independent of  $E$ , only when  $E = 0$ . Then

$$H(B \mid T) = p.$$

The average uncertainty is small, but conditional on the rare event  $E = 1$ , the bit is entirely unknown. If that event is a critical decision, its importance is not measured by its probability alone. Entropy is an average informational quantity, not a measure of the consequences of errors.

These examples distinguish informational equivalence, statistical access, and performance on important cases. A useful translation may need guarantees about all three.

## 4.4 The connection to AI safety

1. **Corresponding information.** Under the reconstruction and Markov conditions, the theorem gives correspondence between specified collections in two models of the same observables.
2. **Concepts of actual agents.** To apply that result, we need to identify suitable variables in actual agents and establish the relevant conditions, including how their observables relate. The theorem does not show that learning procedures produce such models.
3. **A usable translation.** Informational correspondence could support identifying which of an AI’s latents carries something humans care about. We must still find the correspondence with available data and computation, and connect it to the particular human distinction of interest.
4. **Interpretation and specification.** A usable translation could help interpret predictions or express objectives in an AI’s world model. This requires preserving relevant distinctions when circumstances or models change.
5. **Safer systems.** Better interpretation and specification could improve oversight and alignment. Sharing concepts does not entail sharing values, reporting faithfully, or following an intended objective.

The first step is the mathematical result of this lesson. The subsequent steps state a research program and the additional work required to connect it to safety.

## 4.5 Discussion

1. **Choosing the questions.** Consider two agents describing the same physical system, one using local measurements and another using a different collection of measurements. What would have to be established before applying a theorem that assumes the same observables? Distinguish changing value labels from changing which distinctions an observable makes. Identify one restriction on observable choice that seems justified for a concrete application.

2. **What would count as a useful translation?** Suppose two latent tuples  $U, V$  satisfy  $H(U | V) = H(V | U) = 0$ . Specify an additional test that would make this correspondence useful for interpreting a learned model. State what access the test requires: paired samples, interventions, model internals, or a known decoder. Which part of the permutation example would that access resolve?
3. **From correspondence to oversight.** Choose one human distinction that matters for evaluating an AI's behavior. Trace the steps from a correspondence theorem to an oversight procedure that uses that distinction. Name a failure that could occur despite exact informational correspondence, and state what additional evidence would address it.

## 5 Quiz

Take the quiz at [this link](#). The questions are reproduced here because the form cannot typeset mathematics.

### 5.1 Questions

Use base-two logarithms. Throughout,  $S, P, Q$  are independent fair bits and  $X_1 = (S, P)$ ,  $X_2 = (S, Q)$ ,  $X_3 = (P, Q)$ , with latents  $Y_{\{1,2\}} = S$ ,  $Y_{\{1,3\}} = P$  and  $Y_{\{2,3\}} = Q$ .

1. What is  $I(X_1; X_2)$ ?
  - 0
  - 1
  - 2
  - 3
2. Same model. What is  $H(Y_{\{1,2\}} | X_1)$ ?
  - 0
  - 1/2
  - 1
  - 2
3. Now put  $Y_{\{i\}} = X_i$  at each singleton address, with every other slot constant. Which condition fails?
  - the latent variable model condition (LVM)
  - the reconstruction condition
  - the ordered Markov condition
  - none of them fails

4. Complete the two-witness bound  $H(U \mid M) \leq H(U \mid M, P) + H(U \mid M, Q) + ?$ 
  - $I(P; Q)$
  - $I(P; Q \mid M)$
  - $H(P \mid Q)$
  - 0
5. In the correspondence theorem with  $|A| = k$ , suppose every reconstruction error and every Markov defect is at most  $\varepsilon$ . What is the bound on  $H(Y_{\supseteq A} \mid Z_{\supseteq A})$ ?
  - $k\varepsilon$
  - $(k - 1)\varepsilon$
  - $(2k - 1)\varepsilon$
  - $2k\varepsilon$
6. Two corresponding latent collections satisfy  $H(U \mid V) = H(V \mid U) = 0$ . Which of these does that *not* establish?
  - each collection is a function of the other almost everywhere
  - neither collection retains uncertainty about the other
  - an efficient procedure for translating between them
  - the two collections carry the same information

## 6 Further reading

### 6.1 Condensation

The required mathematics is contained in this sheet and the lectures.

- Sam Eisenstat, *Condensation: A Theory of Concepts* (2025). Definitions and exact characterization: Sections 4–5. Correspondence: Theorems 5.15 and 6.8; two-witness inequality: Lemma 6.4 ([Eisenstat, 2025](#)).
- Satya Benson, *Relating Almost Perfect Condensation Conditions and the Conditioned Score*. The score/condition bridge and its reconstruction coverage qualification ([Benson, 2026](#)).
- For the wider statistical context: Cover and Thomas, *Elements of Information Theory* ([Cover and Thomas, 2006](#)); Lauritzen, *Graphical Models* ([Lauritzen, 1996](#)); Shalizi and Crutchfield, *Computational Mechanics: Pattern and Prediction, Structure and Simplicity* (2001) ([Shalizi and Crutchfield, 2001](#)).

## 6.2 Condensation in context

The sources cited above support different parts of the discussion. For the mathematical development, start with Eisenstat’s *Condensation: A Theory of Concepts* (Eisenstat, 2025), particularly the exact and quantitative correspondence results. For the score connection, see *Relating Almost Perfect Condensation Conditions and the Conditioned Score* (Benson, 2026). The score decomposition connects excess conditioned code length to dependence and recoverability defects; the converse requires attention to which observation sets the reconstruction assumptions cover.

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Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3-4): 817–879, 2001. doi: 10.1023/A:1010388907793. URL <https://arxiv.org/abs/cond-mat/9907176>.

Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. In *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, pages 368–377, 1999. URL <https://arxiv.org/abs/physics/0004057>.